HANKEL DETERMINANTS OF DISCRETE MEASURES WITH POWER-LAW ATOMS, AND OF ZETA VALUES

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ABSTRACT. Let $\sigma = \sum_k w_k \delta_{x_k}$ be a discrete measure whose atoms $x_k = f(k)$ accumulate at a point with polynomial spacing, $f'(\xi) \sim -pC\xi^{-p-1}$ for some $p > 0$ and $C > 0$, and whose weights satisfy $\sum_k w_k < \infty$ and $w_k \geq c_1 k^{-s}$. We prove that the Hankel determinants D_n of σ satisfy $\log D_n = -pn^2 \log n + (3p/2 + \log(c(p)C))n^2 + o(n^2)$, where $c(p) = \frac{1}{4}(\sqrt{\pi}\,\Gamma(1/p+1)/\Gamma(1/p+1/2))^p$. Equivalently, a double Cesàro mean of $\log(j^p b_j)$, where the b_j are the off-diagonal Jacobi coefficients of σ, converges to $\log(c(p)C)$. The constant is the value of a logarithmic energy problem on $(0,\infty)$ with an upper constraint of infinite mass near the accumulation point. We solve it explicitly: the minimiser is unique, it is an average of arcsine laws, and its density is a regularised incomplete beta function. As a corollary, for all real $a > 0$ and b with $2a + b > 1$, $\log\det_{1\leq i,j\leq n} \zeta(a(i+j)+b) = -an^2 \log n + (3a/2 + \log c(a))n^2 + o(n^2)$. For positive integers a, b this proves the leading-order behaviour expected by Haynes and Zudilin and identifies the n^2 term. The main theorem and both corollaries are formalised in the Lean 4 proof assistant.

1. INTRODUCTION

1.1. **Setting and main result.** For a positive Borel measure σ on $\mathbb{R}$ with finite moments $m_r = \int x^r \, \mathrm{d}\sigma(x)$, write

$$D_n(\sigma) := \det\big(m_{i+j}\big)_{i,j=0}^{n-1} \qquad (n \geq 1)$$

for its Hankel determinants. We study discrete measures

$$\sigma = \sum_{k\geq 1} w_k \, \delta_{x_k}, \qquad x_k = f(k), \tag{1.1}$$

whose atoms accumulate at a single point with polynomial spacing, the model case being $x_k = Ck^{-p}$. Throughout, $p > 0$ and

$$I(p) := \frac{\sqrt{\pi}}{2}\,\frac{\Gamma(\frac{1}{p}+\frac{1}{2})}{\Gamma(\frac{1}{p}+1)}, \qquad c(p) := \frac{1}{4}\Big(\frac{\pi}{2I(p)}\Big)^p, \qquad \kappa_p := \frac{3p}{2} + \log c(p). \tag{1.2}$$

For instance $I(1) = \pi/4$, $c(1) = \frac{1}{2}$, and $I(2) = 1$, $c(2) = \pi^2/16$.

Theorem 1.1. *Let $p > 0$ and $C > 0$. Let $f \in C^1[1,\infty)$ satisfy*

$$\lim_{\xi\to\infty} \xi^{p+1} f'(\xi) = -pC, \qquad 0 < \inf_{\xi\geq 1} \xi^{p+1}|f'(\xi)| \leq \sup_{\xi\geq 1} \xi^{p+1}|f'(\xi)| < \infty, \tag{1.3}$$

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and let the weights $w_k \ge 0$ *satisfy*

$$W := \sum_{k\ge 1} w_k < \infty \qquad \text{and} \qquad w_k \ge c_1 k^{-s} \quad (k \ge 1) \tag{1.4}$$

for some constants $c_1 > 0$ *and* $s \in \mathbb{R}$. *Then the measure* σ *of* (1.1) *satisfies*

$$\log D_n(\sigma) = -pn^2 \log n + \big(\kappa_p + \log C\big) n^2 + o(n^2) \qquad (n \to \infty). \tag{1.5}$$

The upper bound in the second condition of (1.3) already follows from the limit and the continuity of f'; we state it for symmetry. Under (1.3), f is strictly decreasing with a finite limit L, and $f(\xi) - L \sim C\xi^{-p}$; Hankel determinants are invariant under translation of σ (Lemma 2.2), so L plays no role. The upper bound in (1.5) uses only $\sum_k w_k < \infty$; apart from the finiteness of the moments, the lower bound uses only $w_k \ge c_1 k^{-s}$.

Since σ has infinitely many atoms, it has monic orthogonal polynomials of all degrees and a Jacobi matrix with off-diagonal entries $b_j > 0$ (Lemma 2.4). Theorem 1.1 is equivalent to a doubly averaged statement about these coefficients.

Corollary 1.2. *Under the hypotheses of Theorem 1.1,*

$$\frac{2}{n^2} \sum_{j=1}^{n-1} (n-j) \log\big(j^p b_j\big) \longrightarrow \log\big(c(p)C\big).$$

The constant κ_p is the value of a variational problem, which we solve explicitly. Let

$$\mathcal{A} := \big\{ \mu \text{ positive Borel measure on } (0,\infty) : \ \mu\big((0,\infty)\big) = 1, \ \mu \le \mathrm{Leb} \big\},$$

where $\mu \le \mathrm{Leb}$ means $\mu(B) \le |B|$ for every Borel set B, and for $\mu \in \mathcal{A}$ put

$$\mathcal{J}_p(\mu) := \iint \log\big|u^{-p} - v^{-p}\big| \, \mathrm{d}\mu(u) \, \mathrm{d}\mu(v) \in [-\infty, \infty). \tag{1.6}$$

For $\beta > 0$ let arc_β be the arcsine probability measure on $[0,\beta]$, with density $1/(\pi\sqrt{y(\beta - y)})$. For $y \in [0,1]$ and $a', b' > 0$ let $\mathcal{B}_y(a', b') := B(a', b')^{-1} \int_0^y r^{a'-1}(1-r)^{b'-1} \, \mathrm{d}r$ be the regularised incomplete beta function, where $B(a', b')$ is Euler's beta function.

Theorem 1.3. *Let* $p > 0$ *and* $c = c(p)$. *Let* μ^* *be the probability measure on* $(0,\infty)$

$$\mu^* := \int_0^1 \mathrm{arc}_{4ct^{-p}} \, \mathrm{d}t, \qquad \textit{i.e.} \quad \mu^*(B) = \int_0^1 \mathrm{arc}_{4ct^{-p}}(B) \, \mathrm{d}t,$$

and let μ^*_u *be its image under* $x \mapsto x^{-1/p}$. *Then:*

1. $\mu^*_u \in \mathcal{A}$, *with density* $\rho^*(u) = \mathcal{B}_{\min(1,\, u^{-p}/4c)}\big(\frac{1}{p} + \frac{1}{2}, \frac{1}{2}\big)$. *In particular* $\rho^* = 1$ *on* $(0, u_{\mathrm{sat}}]$, *where* $u_{\mathrm{sat}} := (4c)^{-1/p} = 2I(p)/\pi$, *and* $0 < \rho^* < 1$ *on* $(u_{\mathrm{sat}}, \infty)$.
2. *The function* $U^*(x) := \int \log|x - y| \, \mathrm{d}\mu^*(y)$ *(the negative of the logarithmic potential of* μ^**) satisfies* $U^*(x) = \log c + p$ *for* $0 < x \le 4c$, *and* $\ge \log c + p$ *for all* $x > 0$.
3. $\mathcal{J}_p(\mu^*_u) = \kappa_p$, *and* $\mathcal{J}_p(\mu) < \kappa_p$ *for every* $\mu \in \mathcal{A}$ *with* $\mu \ne \mu^*_u$. *Thus* $\sup_{\mathcal{A}} \mathcal{J}_p = \kappa_p$, *attained only at* μ^*_u.

Applied to the measure $\sum_k k^{-(2a+b)} \delta_{k^{-a}}$, Theorem 1.1 gives the asymptotics of Hankel determinants of zeta values along arithmetic progressions.

Corollary 1.4. *Let $a > 0$ and b be real numbers with $2a + b > 1$. Then*

$$\log \det_{1\le i,j\le n} \zeta\big(a(i+j)+b\big) = -an^2 \log n + \Big(\frac{3a}{2} + \log c(a)\Big) n^2 + o(n^2).$$

In particular, $\log \det_{1\le i,j\le n} \zeta(i+j+b) = -n^2 \log n + (\frac{3}{2} - \log 2) n^2 + o(n^2)$ *for every* $b > -1$, *and* $\log \det_{1\le i,j\le n} \zeta(2(i+j)+b) = -2n^2 \log n + (3 + \log \frac{\pi^2}{16}) n^2 + o(n^2)$ *for every* $b > -3$.

Theorem 1.1 and Corollaries 1.2 and 1.4 have been formalised in Lean 4 with the Mathlib library; see Section 8.

1.2. **Context.** *Hankel determinants of zeta values.* Monien [9] studied the Hankel determinants $H_n^{(r)}[F] = \det(F(i+j+r))$ of Dirichlet series F, gave a multiple-sum formula for them [9, Thm. 2.1], and wrote for $F = \zeta$: "We do not know how to prove the full asymptotic expansions but will describe a method that lets one at least understand the weaker asymptotic form $\log H_n^{(0)}[\zeta] \sim \log H_n^{(1)}[\zeta] \sim -n^2(\log(2n) - 3/2)$." He also reports numerical experiments of D. Zagier consistent with this form. Haynes and Zudilin [6] wrote: "for positive integers a and b, we expect that $\log \det_{1\le i,j\le n} \zeta(a(i+j)+b) = -an^2 \log n + O(n^2)$ as $n \to \infty$", and proved the weaker bound $\log \det_{1\le i,j\le n} \zeta(a(i+j)+b) \le -(a/2)n^2 \log n + O(n^2)$. They add that Monien justified the leading term for $r = 0$ in a personal communication [6, §1]. Coons and Sutherland [4, Thm. 1.3] proved that $H_n^{(r)}[F] > 0$ and $\log H_n^{(r)}[F] < -cn^2$ for large n and r, for nondegenerate Dirichlet series with nonnegative coefficients.

Corollary 1.4 therefore proves the asymptotic formula expected in [6], for all positive integers a, b and more generally for real $a > 0$ and $2a + b > 1$, and it identifies the n^2 term. For $a = 1$ the constant $\frac{3}{2} - \log 2$ agrees with the form reported by Monien (Remark 7.1). For $a = 2$ and $b \in \{-2, 0, 2\}$ the determinants are known in closed form, by results of Baricz and Štampach [1] on Hankel determinants of Rayleigh functions; their asymptotics are consistent with Corollary 1.4 (Remark 7.2). Apart from these closed-form cases, we have not found in the literature a published proof of the leading term $-an^2 \log n$ (for $a = 1$, $r = 0$ see the remark of [6] quoted above), nor of the n^2 term $\frac{3a}{2} + \log c(a)$.

The equilibrium problem. Since $-\mathcal{J}_p(\mu)$ is the logarithmic energy of the image measure $g_*\mu$, where $g(u) = u^{-p}$, the maximisation of $\mathcal{J}_p$ over $\mathcal{A}$ is a logarithmic energy minimisation with an upper constraint on the measure. Problems of this kind were introduced by Rakhmanov [10]. He characterised the constrained equilibrium measure on $[-1, 1]$ for a constraint σ of mass > 1 with continuous potential [10, Thm. 3], and used it to find the limiting zero distribution of polynomials that are extremal for discrete norms on N points as $n/N \to c \in (0, 1)$, in particular of the discrete Chebyshev polynomials [10, Thms. 1 and 2]. Dragnev and Saff added an external field; for a finite constraint measure on a closed set they proved existence, uniqueness, a characterisation and a saturation property of the constrained minimiser [5, Thms. 2.1 and 2.6], with applications to discrete L_p-extremal polynomials on finite intervals [5, Thm. 3.3]. Kuijlaars and Van Assche [8] treated discrete sets $t_k \sim \beta k^{1/\alpha}$ without finite accumulation points and weights $\exp(-\gamma x^\alpha)$. The constraint then has density $\alpha\beta^{-\alpha} x^{\alpha-1}$ on $[0, \infty)$, the weight produces the external field γx^α, and the extremal measure is the Mellin convolution of the beta density $\alpha t^{\alpha-1}$ on $[0, 1]$ with an arcsine density [8, Thms. 2.1 and 2.2]. Beckermann [2, Thm. 1.3] weakened the separation and growth hypotheses of these results and extended them

to complex sets. Van Assche and Van Baelen [13, Thm. 2.2] considered the geometric lattice $\{q^k\}$, which accumulates at 0, with weights that produce no external field; there the constraint has infinite mass near 0, and the problem is posed for circularly symmetric measures in the unit disc after the scaling $x \mapsto x^{1/n}$.

In our problem the atoms accumulate at a finite point with polynomial spacing, there is no external field, and in the variable $x = u^{-p}$ the constraint has infinite mass near the accumulation point. None of the results above covers this case, and we use none of them; Section 3 is self-contained. The maximiser has the same structure as in [8]: μ^* is the law of SY, where Y has the law $\mathrm{arc}_{4c(p)}$ and $S = T^{-p}$ with T uniform on $(0,1)$ and independent of Y. We have not found the problem $\sup_{\mathcal{A}} \mathcal{J}_p$ or its solution in the works cited here.

For $p = 1$, Monien [9, §3] set up this maximisation heuristically, for densities $0 < \rho \le 1$ on $[0,\infty)$, assuming that a continuous maximiser exists; the solution of the stationarity condition in his Theorem 3.1 coincides with ρ^* (Remark 3.5). His argument checks only stationarity and contains no upper bound or discretisation step. Theorem 1.3 proves optimality and uniqueness for every $p > 0$, and Sections 4–5 supply the passage to the determinants.

Averaged arcsine laws. If the off-diagonal and diagonal recurrence coefficients of orthonormal polynomials $p_{n,N}$ tend to continuous functions $\phi(t) \ge 0$ and $\psi(t)$ as $n/N \to t > 0$, the zeros of $p_{n,N}$ are asymptotically distributed as the average over $r \in (0,t)$ of the arcsine laws of $[\psi(r) - 2\phi(r), \psi(r) + 2\phi(r)]$ [7, Thm. 1.4]. For $\phi(r) = cr^{-p}$, $\psi(r) = 2cr^{-p}$ and $t = 1$ this is μ^*. Heuristically, therefore, μ^* is the zero distribution, after scaling by n^p, that the measures of Theorem 1.1 would have if $a_n \sim 2b_n \sim 2c(p)Cn^{-p}$ pointwise; the saturation condition in Theorem 1.3(i) is exactly the equation that fixes $c = c(p)$. This heuristic is not used in the proofs, and Corollary 1.2 concerns double Cesàro means only. Pointwise asymptotics of the recurrence coefficients, with a rate, are the subject of the companion paper [11], which uses Riemann–Hilbert methods for weights $k^{-s}\ell(k)$ with ℓ regular; the companion paper [12] shows that they can fail for weights $k^{-s}e^{\varepsilon_k}$ with $|\varepsilon_k| \le \varepsilon$, to which Theorem 1.1 applies.

1.3. **Method.** By Heine's formula (Lemma 2.1), D_n is a sum over n-subsets S of atoms of $\prod_{k\in S} w_k$ times a squared Vandermonde product. For $x_k = k^{-p}$ and $S = \{k_1 < \cdots < k_n\}$, the Vandermonde product is $n^{-pn(n-1)} \prod_{i\ne j} |g(u_i) - g(u_j)|$ with $g(u) = u^{-p}$ and $u_i = k_i/n$, and the empirical measure $\frac{1}{n}\sum \delta_{u_i}$ has at most one atom in each interval of length $1/n$, a discrete form of the constraint $\mu \le \mathrm{Leb}$. The upper bound (Section 4) compares point configurations with step densities (Lemma 4.1) and removes atoms beyond Mn by a tail estimate; the lower bound (Section 5) evaluates a single term of the Heine sum at the quantiles of a truncation of μ^*_u. Both bounds meet at $\sup_{\mathcal{A}} \mathcal{J}_p$, which Theorem 1.3 identifies. General regularly spaced atoms are reduced to $x_k = Ck^{-p}$ by a mean-value comparison (Lemma 2.3).

The only potential-theoretic input beyond elementary calculus is the positivity of the logarithmic energy of signed measures of total mass zero (Lemma 3.3), a special case of [3, Lemma 2.1] for which we give a short Fourier-analytic proof. It is used only for optimality and uniqueness in Theorem 1.3(iii), which enters the upper bound through $\sup_{\mathcal{A}} \mathcal{J}_p \le \kappa_p$; the lower bound needs only Theorem 1.3(i) and the value $\mathcal{J}_p(\mu^*_u) = \kappa_p$.

1.4. **Organisation and notation.** Section 2 collects the facts about Hankel determinants that we use. Section 3 proves Theorem 1.3; it uses nothing from the later sections and can be read on its own. Sections 4 and 5 prove the upper and lower bounds for $x_k = k^{-p}$, and Section 6 completes the proofs of Theorem 1.1 and Corollary 1.2. Sections 7, 8 and 9 treat zeta values, the formal verification and numerical illustrations.

Throughout, $p > 0$ is fixed and $g(u) := u^{-p}$ on $(0, \infty)$. Constants denoted C_M, C_U, ... depend only on p and on the indicated parameters, and may change from line to line. We use the convention $0 \log 0 = 0$.

2. Hankel determinants of discrete measures

This section collects the facts about Hankel determinants that the proofs use. Heine's formula writes D_n as a sum over n-point subsets of atoms (Lemma 2.1); its consequences include monotonicity in the weights and a comparison lemma that reduces general regularly spaced atoms to $x_k = Ck^{-p}$ (Lemma 2.3). Lemma 2.4 links D_n with the Jacobi coefficients, which gives Corollary 1.2.

In this section $\sigma = \sum_{k \geq 1} w_k \delta_{x_k}$ with $w_k \geq 0$, $\sum_k w_k < \infty$ and $\sup_k |x_k| < \infty$, so that all moments are finite. For a finite set $S \subset \mathbb{N}$ write

$$w(S) := \prod_{k \in S} w_k, \qquad V_x(S) := \prod_{\substack{k,l \in S \\ k<l}} (x_k - x_l)^2.$$

Lemma 2.1 (Heine's formula)**.** *For every $n \geq 1$,*

$$D_n(\sigma) = \sum_{\substack{S \subset \mathbb{N} \\ |S|=n}} w(S)\, V_x(S).$$

In particular all terms are nonnegative.

Proof. Write $m_{i+j} = \sum_k w_k x_k^i x_k^j$. Expanding the determinant multilinearly in its columns (legitimate because $\sum_k w_k |x_k|^r \leq W X^r$ with $W = \sum_k w_k$ and $X := \sup_k |x_k|$, so all series converge absolutely),

$$D_n = \sum_{k_0, \dots, k_{n-1} \geq 1} \Big(\prod_{j=0}^{n-1} w_{k_j} x_{k_j}^{\,j} \Big) \det\big(x_{k_j}^i\big)_{i,j=0}^{n-1}.$$

The Vandermonde determinant vanishes unless the k_j are distinct. Group the remaining tuples according to the set $S = \{l_0 < \cdots < l_{n-1}\}$ they enumerate, $k_j = l_{\pi(j)}$ with π a permutation. Then $\det(x_{k_j}^i) = \operatorname{sgn}(\pi) \det(x_{l_j}^i)$ and

$$\sum_{\pi} \operatorname{sgn}(\pi) \prod_j x_{l_{\pi(j)}}^{\,j} = \det\big(x_{l_i}^{\,j}\big),$$

so the contribution of S is $w(S) \det(x_{l_j}^i)^2 = w(S) V_x(S)$. □

Lemma 2.2. *Let σ, σ' be measures as above.*

1. *(Monotonicity) If $\sigma' = \sum w'_k \delta_{x_k}$ with $w_k \leq w'_k$ for all k, then $D_n(\sigma) \leq D_n(\sigma')$.*
2. *(Translation) $D_n(\sum w_k \delta_{x_k + t}) = D_n(\sigma)$ for every $t \in \mathbb{R}$.*
3. *(Scaling) $D_n(\sum w_k \delta_{\lambda x_k}) = \lambda^{n(n-1)} D_n(\sigma)$ for $\lambda > 0$, and $D_n(\sum \vartheta w_k \delta_{x_k}) = \vartheta^n D_n(\sigma)$ for $\vartheta > 0$.*

(4) *(Positivity) If $w_k > 0$ for infinitely many k with pairwise distinct x_k, then $D_n(\sigma) > 0$ for all n.*

Proof. All four statements follow from Lemma 2.1: its terms are nonnegative and increasing in each w_k; $V_x(S)$ depends only on differences; $V_{\lambda x}(S) = \lambda^{n(n-1)} V_x(S)$ and $(\vartheta w)(S) = \vartheta^n w(S)$; and under the hypothesis of (iv) some term is positive. □

Lemma 2.3 (Regular spacing comparison)**.** *Let $f, h \in C^1[1,\infty)$ with $f' < 0$ and $h' < 0$ on $[1,\infty)$, both bounded on $[1,\infty)$, and suppose that $\rho := f'/h'$ satisfies $R^{-1} \le \rho \le R$ on $[1,\infty)$ for some $R \ge 1$ and $\rho(\xi) \to 1$ as $\xi \to \infty$. Put $\eta(m) := \sup_{\xi \ge m} |\log \rho(\xi)|$. Then for all weights $w_k \ge 0$ with $\sum w_k < \infty$ and all $n \ge 1$,*

$$\Big|\log D_n\Big(\sum_k w_k \delta_{f(k)}\Big) - \log D_n\Big(\sum_k w_k \delta_{h(k)}\Big)\Big| \le 2 \sum_{i=1}^{n-1} (n-i)\,\eta(i) = o(n^2),$$

whenever one (hence both) of the determinants is positive.

Proof. Let $S = \{k_1 < \cdots < k_n\}$. For $i < j$, Cauchy's mean value theorem gives $(f(k_i) - f(k_j))/(h(k_i) - h(k_j)) = \rho(\xi)$ for some $\xi \in (k_i, k_j)$. Since $\xi > k_i \ge i$, the ratio of the squared factors lies in $[e^{-2\eta(i)}, e^{2\eta(i)}]$. Multiplying over pairs, $V_f(S)/V_h(S)$ lies in $[e^{-E_n}, e^{E_n}]$ with $E_n = 2\sum_{i<n}(n-i)\eta(i)$, and the bound follows by comparing the Heine sums term by term. Finally η is bounded by $\log R$ and tends to 0, so $E_n = o(n^2)$. □

Lemma 2.4 (Jacobi coefficients)**.** *Suppose $D_n(\sigma) > 0$ for all n. Let P_k be the monic orthogonal polynomials of σ and $h_k := \int P_k^2 \, d\sigma$. Then $D_n = \prod_{k=0}^{n-1} h_k$. Moreover $xP_k = P_{k+1} + a_k P_k + b_k^2 P_{k-1}$ for $k \ge 1$, with $a_k \in \mathbb{R}$ and $b_k := (h_k/h_{k-1})^{1/2} > 0$, and*

$$D_n(\sigma) = m_0^n \prod_{j=1}^{n-1} b_j^{2(n-j)}.$$

The numbers b_k are the off-diagonal entries of the Jacobi matrix of σ (the coefficients of the three-term recurrence for the orthonormal polynomials $P_k/\sqrt{h_k}$).

Proof. The Hankel matrix $H_n = (m_{i+j})_{i,j<n}$ is the Gram matrix of $1, x, \ldots, x^{n-1}$ in $L^2(\sigma)$. Writing $(P_0, \ldots, P_{n-1})^T = L(1, \ldots, x^{n-1})^T$ with L unit lower triangular gives $LH_nL^T = \operatorname{diag}(h_0, \ldots, h_{n-1})$, so $D_n = \prod_{k<n} h_k$. The polynomial $xP_k - P_{k+1}$ has degree $\le k$, and it is orthogonal to P_l for $l \le k-2$; its P_{k-1}-coefficient is

$$\frac{1}{h_{k-1}} \int xP_kP_{k-1}\, d\sigma = \frac{1}{h_{k-1}} \int P_k\,(xP_{k-1})\, d\sigma = \frac{h_k}{h_{k-1}}.$$

Hence $h_k = m_0 b_1^2 \cdots b_k^2$ and $D_n = \prod_{k=0}^{n-1} m_0 \prod_{j \le k} b_j^2 = m_0^n \prod_{j=1}^{n-1} b_j^{2(n-j)}$. □

Lemma 2.5. $2\sum_{j=1}^{n-1}(n-j)\log j = n^2 \log n - \frac{3}{2}n^2 + O(n \log n)$.

Proof. Let $\phi(x) = (n-x)\log x$ on $[1,n]$. Then $|\phi'(x)| \le \log n + n/x$, so $\big|\sum_{j=1}^{n-1} \phi(j) - \int_1^n \phi\big| \le \sum_{j=1}^{n-1}(\log n + n/j) = O(n \log n)$, and $\int_1^n (n-x)\log x \, dx = \frac{n^2}{2}\log n - \frac{3}{4}n^2 + O(n)$. □

3. The constrained equilibrium problem

This section proves Theorem 1.3. The upper bound of Section 4 needs the value $\sup_{\mathcal{A}} \mathcal{J}_p$, and the lower bound of Section 5 needs the maximiser itself. After a scaling lemma (Lemma 3.1) and the logarithmic potential of the arcsine law (Lemma 3.2), we prove that signed measures of total mass zero have nonnegative logarithmic energy (Lemma 3.3); this gives optimality and uniqueness in Theorem 1.3(iii).

3.1. **Energies.** For finite positive Borel measures μ, ν on $(0,\infty)$ with $\int \log(1+x)\,\mathrm{d}\mu < \infty$ and $\int \log(1+x)\,\mathrm{d}\nu < \infty$ put

$$\mathcal{L}(\mu,\nu) := \iint \log|x-y|\,\mathrm{d}\mu(x)\,\mathrm{d}\nu(y), \qquad \mathcal{L}(\mu) := \mathcal{L}(\mu,\mu).$$

Since $\log|x-y| \le \log(1+x)+\log(1+y)$ for $x,y>0$, the positive part of the integrand is integrable, so $\mathcal{L}(\mu,\nu) \in [-\infty,\infty)$ is well defined. Let

$$\lambda_p(\,\mathrm{d}x) := \tfrac{1}{p}\,x^{-1-1/p}\,\mathrm{d}x \quad \text{on } (0,\infty),$$

the image of Lebesgue measure on $(0,\infty)$ under $g(u)=u^{-p}$. For $\tau>0$ let $\mathcal{A}_\tau$ be the set of positive Borel measures on $(0,\infty)$ of mass τ with $\mu \le \mathrm{Leb}$; thus $\mathcal{A}=\mathcal{A}_1$. For $\mu \in \mathcal{A}_\tau$ the image $\mu^x := g_*\mu$ satisfies $\mu^x \le \lambda_p$, hence $\int \log(1+x)\,\mathrm{d}\mu^x \le \tau\log 2 + \int_1^\infty \log(1+x)\,\mathrm{d}\lambda_p < \infty$, and we define

$$\mathcal{J}_p(\mu) := \mathcal{L}(\mu^x) = \iint \log|g(u)-g(v)|\,\mathrm{d}\mu(u)\,\mathrm{d}\mu(v) \in [-\infty,\infty),$$
$$E_p := \sup_{\mu\in\mathcal{A}} \mathcal{J}_p(\mu).$$

Lemma 3.1 (Scaling)**.** *For a measure μ on $(0,\infty)$ and $\tau>0$ let $\mu_\tau(B) := \tau\mu(B/\tau)$. If μ has mass m and density $\rho \le 1$, then μ_τ has mass τm and density $\rho(\cdot/\tau) \le 1$, and $\mathcal{J}_p(\mu_\tau) = \tau^2(\mathcal{J}_p(\mu) - pm^2\log\tau)$. In particular $\mu \mapsto \mu_\tau$ is a bijection $\mathcal{A} \to \mathcal{A}_\tau$ and $\sup_{\mathcal{A}_\tau}\mathcal{J}_p = \tau^2(E_p - p\log\tau)$.*

Proof. $\mu_\tau(B) = \tau\int_{B/\tau}\rho(u)\,\mathrm{d}u = \int_B \rho(v/\tau)\,\mathrm{d}v$. Since $g(\tau u) = \tau^{-p}g(u)$,

$$\mathcal{J}_p(\mu_\tau) = \tau^2 \iint \log|g(\tau u)-g(\tau v)|\,\mathrm{d}\mu\,\mathrm{d}\mu = \tau^2\big(\mathcal{J}_p(\mu) - pm^2\log\tau\big). \qquad \square$$

3.2. **Arcsine potentials.**

Lemma 3.2. *Let $\beta>0$ and $U_\beta(x) := \int \log|x-y|\,\mathrm{darc}_\beta(y)$. Then $U_\beta(x) = \log(\beta/4)$ for $x \in [0,\beta]$, and U_β is increasing on $[\beta,\infty)$; in particular $U_\beta \ge \log(\beta/4)$ on $[0,\infty)$. Moreover, for every $x \ge 0$,*

$$\int \big|\log|x-y|\big|\,\mathrm{darc}_\beta(y) \le 2\log(1+x+\beta) + \big|\log(\beta/4)\big|. \tag{3.1}$$

Proof. Write $x = \frac{\beta}{2}(1-\cos\varphi)$ and $y = \frac{\beta}{2}(1-\cos\theta)$ with $\theta,\varphi \in [0,\pi]$. Under $y = \frac{\beta}{2}(1-\cos\theta)$, the measure arc_β is the image of $\mathrm{d}\theta/\pi$ on $[0,\pi]$, since $\sqrt{y(\beta-y)} = \frac{\beta}{2}\sin\theta$. Now $\log|x-y| = \log\frac{\beta}{4} + \log|2(\cos\theta-\cos\varphi)|$, and

$$\log|2(\cos\theta-\cos\varphi)| = \log\Big|2\sin\tfrac{\theta+\varphi}{2}\Big| + \log\Big|2\sin\tfrac{\theta-\varphi}{2}\Big| = -\sum_{k\ge1}\frac{2}{k}\cos k\theta\cos k\varphi,$$

using $\log|2\sin(\alpha/2)| = -\sum_{k\ge1}k^{-1}\cos k\alpha$, the Fourier series of an $L^2(0,2\pi)$ function. Each of the two series converges in $L^2([0,\pi],\mathrm{d}\theta)$, hence in L^1, so we may

integrate termwise; since $\int_0^\pi \cos k\theta \, \mathrm{d}\theta = 0$ for $k \geq 1$, we get $U_\beta(x) = \log(\beta/4)$ for $x \in [0, \beta]$. For $x > \beta$, $U'_\beta(x) = \int (x-y)^{-1} \operatorname{darc}_\beta(y) > 0$, and U_β is continuous at β by dominated convergence (near $y = \beta$ the integrand is dominated by a multiple of $|\log r| \, r^{-1/2}$, $r = \beta - y$). For (3.1), write $|t| = 2t^+ - t$: since $|x-y| \leq x+\beta$ on the support, $\int |\log|x-y|| \operatorname{darc}_\beta = 2\int \log^+ |x-y| \operatorname{darc}_\beta - U_\beta(x) \leq 2\log(1+x+\beta) - \log(\beta/4)$. □

3.3. **Positivity of the energy of signed measures.**

Lemma 3.3. *Let ν be a finite signed Borel measure on $\mathbb{R}$ with $\nu(\mathbb{R}) = 0$ and $\iint |\log|x-y|| \, \mathrm{d}|\nu|(x) \, \mathrm{d}|\nu|(y) < \infty$. Then*

$$-\iint \log|x-y| \, \mathrm{d}\nu(x) \, \mathrm{d}\nu(y) = \int_0^\infty \frac{|\hat\nu(\xi)|^2}{\xi} \, \mathrm{d}\xi \ \geq 0,$$

where $\hat\nu(\xi) := \int e^{i\xi x} \, \mathrm{d}\nu(x)$, with equality if and only if $\nu = 0$.

Proof. The idea is to write $-\log|a|$ as a limit of the truncated integrals $K_{\varepsilon,R}(a) := \int_\varepsilon^R (\cos(a\xi) - e^{-\xi}) \, \xi^{-1} \, \mathrm{d}\xi$, defined for $0 < \varepsilon < 1 < R$ and $a \in \mathbb{R} \setminus \{0\}$. For each truncation, Fubini's theorem turns $\iint K_{\varepsilon,R}(x-y) \, \mathrm{d}\nu \, \mathrm{d}\nu$ into $\int_\varepsilon^R |\hat\nu(\xi)|^2 \xi^{-1} \, \mathrm{d}\xi$, because $\nu(\mathbb{R}) = 0$ removes the term $e^{-\xi}$. A bound on $K_{\varepsilon,R}$ that is uniform in ε and R (Step 1) and its pointwise limit (Step 2) allow the passage to the limit on the energy side (Step 3). The equality case is Fourier uniqueness.

Step 1: a uniform bound. We use three elementary estimates: (a) for $1 \leq A < B$, $|\int_A^B \cos t \, \frac{\mathrm{d}t}{t}| \leq 2/A \leq 2$ (integrate by parts); (b) for $0 < A < B \leq 1$, $\int_A^B \cos t \, \frac{\mathrm{d}t}{t} = \log\frac{B}{A} - \theta$ with $0 \leq \theta \leq \int_A^B \frac{t}{2} \, \mathrm{d}t \leq \frac{B^2}{4} \leq \frac{1}{4}$, because $0 \leq 1 - \cos t \leq t^2/2$; (c) $\int_\varepsilon^R e^{-\xi} \frac{\mathrm{d}\xi}{\xi} = \log\frac{1}{\varepsilon} + \theta'$ with $|\theta'| \leq 2$. Substitute $t = |a|\xi$, so that $\int_\varepsilon^R \cos(a\xi) \frac{\mathrm{d}\xi}{\xi} = \int_{\varepsilon|a|}^{R|a|} \cos t \frac{\mathrm{d}t}{t}$. If $\varepsilon|a| < 1 < R|a|$, then $K_{\varepsilon,R}(a) = -\log|a| + O(1)$. If $R|a| \leq 1$, then $K_{\varepsilon,R}(a) = \log R + O(1)$ and $0 < \log R \leq \log(1/|a|)$. If $\varepsilon|a| \geq 1$, then $K_{\varepsilon,R}(a) = -\log(1/\varepsilon) + O(1)$ and $0 < \log(1/\varepsilon) \leq \log|a|$. In all cases

$$|K_{\varepsilon,R}(a)| \leq \big|\log|a|\big| + 6 \qquad (0 < \varepsilon < 1 < R, \ a \neq 0). \tag{3.2}$$

Step 2: the pointwise limit. We have $K_{\varepsilon,R}(a) - K_{\varepsilon,R}(1) = \int_{\varepsilon|a|}^{\varepsilon} \cos t \frac{\mathrm{d}t}{t} + \int_R^{R|a|} \cos t \frac{\mathrm{d}t}{t}$. As $\varepsilon \to 0$ the first integral tends to $\log(1/|a|)$, by (b); as $R \to \infty$ the second tends to 0, by (a). Also $K_{\varepsilon,R}(1) \to k_0 := \int_0^\infty (\cos\xi - e^{-\xi}) \xi^{-1} \, \mathrm{d}\xi$, which converges. Hence $K_{\varepsilon,R}(a) \to k_0 - \log|a|$. (In fact $k_0 = 0$, but we do not need this, because $\nu(\mathbb{R}) = 0$.) All limits as $\varepsilon \to 0$, $R \to \infty$ are taken along arbitrary sequences $\varepsilon_m \downarrow 0$, $R_m \uparrow \infty$.

Step 3. The integrability hypothesis forces $|\nu| \otimes |\nu|$ to give zero mass to the diagonal $\{x = y\}$. By (3.2) and dominated convergence along (ε_m, R_m) (with respect to $|\nu| \otimes |\nu|$, dominating function $|\log|x-y|| + 6$), and since $\nu(\mathbb{R}) = 0$,

$$\begin{aligned}\iint K_{\varepsilon,R}(x-y) \, \mathrm{d}\nu \, \mathrm{d}\nu &\longrightarrow k_0 \, \nu(\mathbb{R})^2 - \iint \log|x-y| \, \mathrm{d}\nu \, \mathrm{d}\nu \\ &= -\iint \log|x-y| \, \mathrm{d}\nu \, \mathrm{d}\nu.\end{aligned}$$

On the other hand, the integrand $(\xi,x,y)\mapsto(\cos((x-y)\xi)-e^{-\xi})/\xi$ is bounded on $[\varepsilon,R]\times\mathbb{R}^2$, so by Fubini, $\iint\cos((x-y)\xi)\,\mathrm{d}\nu\,\mathrm{d}\nu=|\hat\nu(\xi)|^2$ and $\nu(\mathbb{R})=0$,

$$\iint K_{\varepsilon,R}(x-y)\,\mathrm{d}\nu\,\mathrm{d}\nu=\int_\varepsilon^R\frac{|\hat\nu(\xi)|^2}{\xi}\,\mathrm{d}\xi.$$

Letting $\varepsilon\to0$, $R\to\infty$ (monotone convergence on the right) gives the identity.

Equality. If the integral vanishes, then $\hat\nu=0$ on $(0,\infty)$ by continuity, hence on $\mathbb{R}$, because $\hat\nu(-\xi)=\overline{\hat\nu(\xi)}$ and $\hat\nu(0)=\nu(\mathbb{R})=0$. Then $\nu=0$: for the Gaussian $G_t(x)=(2\pi t)^{-1/2}e^{-x^2/2t}$ we have $G_t(x)=\frac{1}{2\pi}\int e^{-t\xi^2/2}e^{-i\xi x}\,\mathrm{d}\xi$, so by Fubini $(G_t*\nu)(x)=\frac{1}{2\pi}\int e^{-t\xi^2/2}e^{-i\xi x}\hat\nu(\xi)\,\mathrm{d}\xi=0$ for all x and $t>0$; for bounded continuous ψ, $\int\psi\,\mathrm{d}\nu=\lim_{t\to0}\int(G_t*\psi)\,\mathrm{d}\nu=\lim_{t\to0}\int\psi\,(G_t*\nu)\,\mathrm{d}x=0$ by dominated convergence, so $\nu=0$. □

Remark 3.4. The positivity statement of Lemma 3.3 (without the Fourier identity) is a special case of [3, Lemma 2.1]. That lemma applies to positive measures of equal finite mass with finite energy and $\int\log(1+|t|)\,\mathrm{d}\mu(t)<\infty$ [3, (1.12)]; its energy $I(\mu)=\iint\log|x-y|^{-1}\,\mathrm{d}\mu\,\mathrm{d}\mu$ is not our constant $I(p)$. The positive and negative parts of ν in Lemma 3.3 satisfy these conditions; for the logarithmic moment, fix y with $\int|\log|x-y||\,\mathrm{d}|\nu|(x)<\infty$ and use $\log(1+|x|)\le\log(1+|y|)+\log2+\log^+|x-y|$.

3.4. **The explicit maximiser.** Throughout this subsection $c=c(p)$ and $\beta_t:=4ct^{-p}$ for $t\in(0,1]$. Thus μ^* is the mixture $\mu^*(B)=\int_0^1\operatorname{arc}_{\beta_t}(B)\,\mathrm{d}t$ for Borel sets B. Put $U^*(x):=\int\log|x-y|\,\mathrm{d}\mu^*(y)$.

Proof of Theorem 1.3(i). Fix $x>0$ and let $t_x:=(4c/x)^{1/p}$, so that $x<\beta_t$ if and only if $t<t_x$. Then

$$\frac{\mathrm{d}\mu^*}{\mathrm{d}x}=\frac1\pi\int_0^{\min(1,t_x)}\frac{\mathrm{d}t}{\sqrt{x(\beta_t-x)}}=\frac{t_x}{\pi x}\int_0^{\min(1,1/t_x)}\frac{\mathrm{d}r}{\sqrt{r^{-p}-1}},$$

by the substitution $t=t_xr$, for which $\beta_t=xr^{-p}$. The substitution $v=r^p$ gives, for $0\le y\le1$,

$$\int_0^{y^{1/p}}\frac{\mathrm{d}r}{\sqrt{r^{-p}-1}}=\frac1p\int_0^y v^{\frac1p-\frac12}(1-v)^{-\frac12}\,\mathrm{d}v=\frac1pB\big(\tfrac1p+\tfrac12,\tfrac12\big)\mathcal{B}_y\big(\tfrac1p+\tfrac12,\tfrac12\big),$$

and $B(\frac1p+\frac12,\frac12)=\sqrt\pi\,\Gamma(\frac1p+\frac12)/\Gamma(\frac1p+1)=2I(p)$. With $y=\min(1,t_x^{-p})=\min(1,x/4c)$ we obtain

$$\begin{aligned}\frac{\mathrm{d}\mu^*}{\mathrm{d}x}&=\frac{(4c)^{1/p}x^{-1-1/p}}{\pi}\cdot\frac{2I(p)}{p}\cdot\mathcal{B}_{\min(1,x/4c)}\big(\tfrac1p+\tfrac12,\tfrac12\big)\\&=\mathcal{B}_{\min(1,x/4c)}\big(\tfrac1p+\tfrac12,\tfrac12\big)\cdot\frac{\mathrm{d}\lambda_p}{\mathrm{d}x},\end{aligned}$$

because $(4c)^{1/p}=\pi/(2I(p))$ by the definition of $c=c(p)$. Thus $\mu^*=\lambda_p$ on $[4c,\infty)$ and $\mu^*<\lambda_p$ (in density) on $(0,4c)$. Pulling back by $x=u^{-p}$, which maps λ_p to Lebesgue measure, gives the density ρ^* of μ^*_u; ρ^* is continuous and positive on $(0,\infty)$, equal to 1 exactly on $(0,u_{\mathrm{sat}}]$, and $u_{\mathrm{sat}}=(4c)^{-1/p}=2I(p)/\pi$. The total mass is $\int_0^1\mathrm{d}t=1$. Hence $\mu^*_u\in\mathcal{A}$. □

Proof of Theorem 1.3(ii). By (3.1), for fixed $x>0$, $\int_0^1\int|\log|x-y||\,\mathrm{d}\operatorname{arc}_{\beta_t}(y)\,\mathrm{d}t\le\int_0^1\big(2\log(1+x+\beta_t)+|\log(\beta_t/4)|\big)\,\mathrm{d}t<\infty$, since $\log\beta_t=\log4c-p\log t$. By

Fubini, $U^*(x) = \int_0^1 U_{\beta_t}(x)\,\mathrm{d}t$. If $x \le 4c = \beta_1$, then $x \le \beta_t$ for all $t \in (0,1]$, and Lemma 3.2 gives $U^*(x) = \int_0^1 \log(ct^{-p})\,\mathrm{d}t = \log c + p$. For all $x > 0$, Lemma 3.2 gives $U_{\beta_t}(x) \ge \log(\beta_t/4)$, hence $U^*(x) \ge \log c + p$. □

Proof of Theorem 1.3(iii). The value. For $0 < r \le t \le 1$ we have $\operatorname{supp}\operatorname{arc}_{\beta_t} \subset [0,\beta_t] \subset [0,\beta_r]$, so by Lemma 3.2

$$\iint \log|x-y|\,\operatorname{darc}_{\beta_r}(x)\,\operatorname{darc}_{\beta_t}(y) = \int U_{\beta_r}\,\operatorname{darc}_{\beta_t} = \log\frac{\beta_r}{4} = \log c - p\log\min(r,t).$$

By (3.1), $\iint |\log|x-y||\,\operatorname{darc}_{\beta_r}\operatorname{darc}_{\beta_t} \le 2\log(1+\beta_r+\beta_t) + |\log(\beta_t/4)|$, which is integrable over $(r,t) \in (0,1]^2$. Hence $\mathcal{L}(\mu^*)$ converges absolutely, and by Fubini

$$\mathcal{J}_p(\mu_u^*) = \mathcal{L}(\mu^*) = \int_0^1\int_0^1 \big(\log c - p\log\min(r,t)\big)\,\mathrm{d}r\,\mathrm{d}t = \log c - p\int_0^1 2(1-m)\log m\,\mathrm{d}m = \log c + \frac{3p}{2} = \kappa_p.$$

Optimality. Let $\mu \in \mathcal{A}$. If $\mathcal{J}_p(\mu) = -\infty$ there is nothing to prove, so assume $\mathcal{J}_p(\mu) > -\infty$ and let $\mu^x = g_*\mu \le \lambda_p$. We first check that all energies below converge absolutely.

- $\mathcal{L}(\mu^x)$: the positive part of the integrand is integrable (Section 3.1), and the negative part is integrable because $\mathcal{L}(\mu^x) > -\infty$.
- $\mathcal{L}(\mu^*)$: shown above.
- $\mathcal{L}(\mu^x,\mu^*)$: using $|t| = 2t^+ - t$, $\log^+|x-y| \le \log(1+x) + \log(1+y)$ and part (ii),

$$\iint \big|\log|x-y|\big|\,\mathrm{d}\mu^*(y)\,\mathrm{d}\mu^x(x) \le \int\Big(2\log(1+x) + 2\int\log(1+y)\,\mathrm{d}\mu^*(y) - (\log c + p)\Big)\,\mathrm{d}\mu^x(x) < \infty,$$

since $\mu^* \le \lambda_p$ and $\mu^x \le \lambda_p$ have finite logarithmic moments and mass 1.

Let $\nu := \mu^x - \mu^*$, a finite signed measure on $(0,\infty)$ with $\nu(\mathbb{R}) = 0$. Since $|\nu| \le \mu^x + \mu^*$, the three bounds give $\iint |\log|x-y||\,\mathrm{d}|\nu|\,\mathrm{d}|\nu| < \infty$. Expanding bilinearly (all terms are finite),

$$\mathcal{J}_p(\mu) = \mathcal{L}(\mu^x) = \mathcal{L}(\mu^*) + 2\int U^*\,\mathrm{d}\nu + \iint \log|x-y|\,\mathrm{d}\nu\,\mathrm{d}\nu.$$

By Lemma 3.3 the last term is ≤ 0, with equality only if $\nu = 0$. For the middle term, ν has mass 0, so $\int U^*\,\mathrm{d}\nu = \int (U^* - \log c - p)\,\mathrm{d}\nu$. By (ii) the integrand vanishes on $(0,4c]$ and is ≥ 0 on $(4c,\infty)$, where $\nu = \mu^x - \lambda_p \le 0$ by (i). Hence $\int U^*\,\mathrm{d}\nu \le 0$, and $\mathcal{J}_p(\mu) \le \mathcal{L}(\mu^*) = \kappa_p$, with equality only if $\nu = 0$, that is, $\mu = \mu_u^*$. □

Remark 3.5. For $p = 1$, Monien's solution of the stationarity condition [9, Thm. 3.1] is

$$\rho(x) = 1 \quad (0 \le x \le \tfrac12), \qquad \rho(x) = \tfrac{2}{\pi}\Big(\arctan\tfrac{1}{\sqrt{2x-1}} - \tfrac{\sqrt{2x-1}}{2x}\Big) \quad (x > \tfrac12).$$

We have $c = \frac12$, $u_{\mathrm{sat}} = \frac12$ and $\rho^*(u) = \mathcal{B}_{\min(1,1/2u)}(\frac32,\frac12)$, and $\rho^* = \rho$. Both equal 1 on $(0,\frac12]$. For $u > \frac12$ put $y = 1/(2u)$; since $B(\frac32,\frac12) = \frac{\pi}{2}$, the substitution $v = \sin^2\phi$

gives

$$\mathcal{B}_y\big(\tfrac{3}{2},\tfrac{1}{2}\big)=\frac{2}{\pi}\int_0^y\sqrt{\frac{v}{1-v}}\,\mathrm{d}v=\frac{2}{\pi}\big(\phi_y-\sin\phi_y\cos\phi_y\big),\qquad \sin\phi_y=\sqrt{y},$$

and $\phi_y=\arctan(1/\sqrt{2u-1})$, $\sin\phi_y\cos\phi_y=\sqrt{2u-1}/(2u)$.

4. Discretisation and the upper bound

This section proves the upper bound in Theorem 1.1 for $x_k=k^{-p}$, with the constant $E_p=\sup_{\mathcal{A}}\mathcal{J}_p$. Configurations of atoms inside $[1,Mn]$ are compared with step densities in $\mathcal{A}_\tau$ (Lemma 4.1); atoms beyond Mn are removed by a direct tail estimate (Theorem 4.3).

In Sections 4 and 5 we write, for a finite $S=\{k_1<\cdots<k_N\}\subset\mathbb{N}$,

$$V(S):=\prod_{i<j}\big(k_i^{-p}-k_j^{-p}\big)^2,\qquad u_i:=k_i/n.$$

Since $k^{-p}=n^{-p}g(k/n)$,

$$\log V(S)=-pN(N-1)\log n+\sum_{i\neq j}\log|g(u_i)-g(u_j)|. \tag{4.1}$$

Lemma 4.1 (Discretisation)**.** *Let $M\geq 1$. There is C_M such that for all $n\geq 2$ and all $S=\{k_1<\cdots<k_N\}\subset\mathbb{N}\cap[1,Mn]$ with $N\leq n$,*

$$\sum_{i\neq j}\log|g(u_i)-g(u_j)|\leq n^2\mathcal{J}_p(\bar\mu_S)+C_Mn\log n,$$

where $\bar\mu_S\in\mathcal{A}_{N/n}$ has density 1 on $\bigcup_{k\in S}A_k$, $A_k:=[(k-1)/n,k/n]$.

Proof. For cells A,A' write $\langle F\rangle_{A\times A'}:=n^2\iint_{A\times A'}F(u,v)\,\mathrm{d}u\,\mathrm{d}v$ for the average of $F(u,v):=\log|g(u)-g(v)|$. Then $n^2\mathcal{J}_p(\bar\mu_S)=\sum_{i\neq j}\langle F\rangle_{A_{k_i}\times A_{k_j}}+\sum_i\langle F\rangle_{A_{k_i}^2}$. Recall that $|g'(u)|=pu^{-p-1}$ is decreasing.

Non-adjacent pairs. Let $k<l$ with $l\geq k+2$. For $u\in A_k$, $v\in A_l$ we have $g(u)-g(v)\geq g(k/n)-g((l-1)/n)=:D>0$. With $\Delta:=g((l-1)/n)-g(l/n)\leq|g'((l-1)/n)|/n$ and $D\geq\frac{l-1-k}{n}|g'((l-1)/n)|$ (mean value theorem and monotonicity of $|g'|$),

$$\log\big(g(k/n)-g(l/n)\big)=\log(D+\Delta)\leq\log D+\frac{\Delta}{D}\leq\langle F\rangle_{A_k\times A_l}+\frac{1}{l-k-1}.$$

Adjacent pairs. Let $l=k+1$. The point value is $\log(g(k/n)-g((k+1)/n))\leq\log(|g'(k/n)|/n)$. For $u\in A_k$, $v\in A_{k+1}$, $g(u)-g(v)\geq|g'((k+1)/n)|(v-u)$, so $\langle F\rangle_{A_k\times A_{k+1}}\geq\log|g'((k+1)/n)|-\log n+\vartheta_0$ with $\vartheta_0:=\int_0^1\int_1^2\log(y-x)\,\mathrm{d}y\,\mathrm{d}x=2\log 2-\frac{3}{2}$. Since $|g'(k/n)|/|g'((k+1)/n)|=(1+1/k)^{p+1}\leq 2^{p+1}$, the point value exceeds the average by at most $(p+1)\log 2-\vartheta_0$.

Diagonal cells. For $u,v\in A_k$, $|g(u)-g(v)|\geq|g'(k/n)||u-v|$, so $\langle F\rangle_{A_k^2}\geq\log|g'(k/n)|-\log n-\frac{3}{2}$, using $\iint_{[0,1]^2}\log|x-y|=-\frac{3}{2}$. For $k\leq Mn$, $|g'(k/n)|\geq pM^{-p-1}$.

Summation. For fixed i, the partners $j>i$ with $k_j\geq k_i+2$ have distinct values $k_j-k_i-1\geq 1$, so $\sum_j 1/(k_j-k_i-1)\leq 1+\log n$; there is at most one adjacent

partner. Counting ordered pairs,

$$\sum_{i\neq j}\log|g(u_i)-g(u_j)| \le n^2\mathcal{J}_p(\bar\mu_S)+2N\big(1+\log n+(p+1)\log 2+|\vartheta_0|\big) + N\big(\log n+\tfrac{3}{2}+(p+1)\log M+|\log p|\big),$$

which is of the stated form since $N\le n$. □

The diagonal estimate is the only place where $S\subset[1,Mn]$ is used: an isolated point $k\gg n$ makes $\mathcal{J}_p(\bar\mu_S)$ very negative while contributing little to the point sum. Atoms beyond Mn are therefore treated separately in Theorem 4.3.

Corollary 4.2. *For $M\ge 1$ there is C'_M such that for $n\ge 2$ and $S\subset\mathbb{N}\cap[1,Mn]$ with $|S|=N\le n$,*

$$\log V(S)\le -pN^2\log N+E_pN^2+C'_Mn\log n.$$

Proof. For $N=0$ both sides are trivial. Otherwise, by (4.1), Lemma 4.1 and Lemma 3.1 with $\tau=N/n$,

$$\log V(S)\le -pN(N-1)\log n+N^2\big(E_p-p\log(N/n)\big)+C_Mn\log n,$$

and $-pN(N-1)\log n-pN^2\log(N/n)=-pN^2\log N+pN\log n$. □

Theorem 4.3 (Upper bound)**.** *Let $w_k\ge 0$ with $W=\sum_k w_k<\infty$ and $\sigma_0:=\sum_k w_k\delta_{k^{-p}}$. Then*

$$\limsup_{n\to\infty}\frac{1}{n^2}\Big(\log D_n(\sigma_0)+pn^2\log n\Big)\le E_p.$$

Proof. We split each n-subset S of atoms into a head H inside $[1,Mn]$ and a tail T beyond Mn. The head is controlled by Corollary 4.2, which produces the energy E_p. The tail and the cross factors are bounded crudely: all tail atoms lie in $(0,(Mn)^{-p})$, and a tail atom is closer than a^{-p} to the head atom a^{-p}. Optimising over the proportion t of tail atoms leaves an error $A^2/(4p\log M)$, which vanishes as $M\to\infty$.

Fix $M\ge 2$. Each n-set $S\subset\mathbb{N}$ splits uniquely as $S=H\cup T$ with $H:=S\cap[1,Mn]$ and $T:=S\cap(Mn,\infty)$; let $m:=|T|$ and $N:=n-m=|H|$. Then $V(S)=V(H)V(T)\prod_{a\in H,b\in T}(a^{-p}-b^{-p})^2$.

Cross pairs. For $a\in H$ and $b\in T$, $0<a^{-p}-b^{-p}<a^{-p}$. Since H consists of N distinct positive integers, $\prod_{a\in H}a\ge N!$, so the cross product is at most $(N!)^{-2pm}$.

Tail. All atoms b^{-p} with $b\in T$ lie in the interval $(0,(Mn)^{-p})$; hence

$$V(T)\le (Mn)^{-pm(m-1)},\qquad \sum_{|T|=m}w(T)\le W^m.$$

Head. $\sum_{|H|=N}w(H)V(H)\le W^N\sup_H V(H)$, and Corollary 4.2 bounds $\sup_H V(H)$.

By Lemma 2.1, $D_n(\sigma_0)\le\sum_{m=0}^{n}T_m$, where

$$\log T_m\le n\log^+W-2pm\log N!-pN^2\log N+E_pN^2 - pm(m-1)\log(Mn)+C'_Mn\log n.$$

Recall that $E_p=\kappa_p$ is finite (Theorem 1.3(iii)). Put $t:=m/n$. If $N\ge 1$, use $\log N!\ge N\log N-N$, $\log N=\log n+\log(1-t)$ and $\log(Mn)=\log n+\log M$.

The coefficient of $\log n$ is $-p(2mN+N^2+m^2) = -pn^2$, and the remaining terms give

$$\log T_m + pn^2 \log n - E_p n^2 \le n^2 \phi_M(t) + O_M(n \log n),$$
$$\phi_M(t) := -p(1-t)(1+t)\log(1-t) + 2pt(1-t) - E_p\, t(2-t) - pt^2 \log M.$$

Since $-(1-t)\log(1-t) \le t$ on $[0,1)$, we get $\phi_M(t) \le p(1+t)t + 2pt + 2|E_p|t - pt^2 \log M \le At - pt^2 \log M$ with $A := 4p + 2|E_p|$. If $N = 0$ ($t = 1$), then directly $\log T_n \le n \log^+ W - pn(n-1)\log(Mn) + C'_M n \log n$, and $-p \log M - E_p \le A - p \log M$, so the same bound holds. The $O_M(n \log n)$ terms are uniform in m, and $\max_{t\in[0,1]}(At - pt^2 \log M) \le A^2/(4p \log M)$. Summing the $n+1$ terms,

$$\limsup_{n\to\infty} \frac{1}{n^2}\Big(\log D_n(\sigma_0) + pn^2 \log n\Big) \le E_p + \frac{A^2}{4p \log M}.$$

Let $M \to \infty$. □

5. The lower bound

This section proves the matching lower bound $\liminf \ge \kappa_p$ for $x_k = k^{-p}$ and weights $w_k \ge c_1 k^{-s}$. By Heine's formula it suffices to find one n-subset of atoms whose term is large. We take the atoms nearest to the quantiles of a compactly supported approximation of μ_u^*, and compare the discrete energy with the continuous one block by block.

Lemma 5.1. *Let φ be decreasing on $(0,\infty)$ and integrable near 0, and let $\lambda \le$ Leb be a measure on $(0,\infty)$ of mass $\delta > 0$ with $\int |\varphi|\,\mathrm{d}\lambda < \infty$. Then $\int \varphi\,\mathrm{d}\lambda \le \int_0^\delta \varphi(u)\,\mathrm{d}u$.*

Proof. $\int \varphi\,\mathrm{d}\lambda - \int_0^\delta \varphi = \int_{(0,\delta]} \varphi\,\mathrm{d}(\lambda - \mathrm{Leb}) + \int_{(\delta,\infty)} \varphi\,\mathrm{d}\lambda \le \varphi(\delta)\big(\lambda((0,\delta]) - \delta\big) + \varphi(\delta)\lambda((\delta,\infty)) = 0$, because $\lambda - \mathrm{Leb} \le 0$ and $\varphi \ge \varphi(\delta)$ on $(0,\delta]$, and $\varphi \le \varphi(\delta)$ on (δ,∞). □

Theorem 5.2 (Lower bound)**.** *Let $w_k \ge c_1 k^{-s}$ for all k, with $c_1 > 0$, $s \in \mathbb{R}$, and let $\sigma_0 = \sum_k w_k \delta_{k^{-p}}$ (with $\sum w_k < \infty$, so that the moments are finite). Then*

$$\liminf_{n\to\infty} \frac{1}{n^2}\Big(\log D_n(\sigma_0) + pn^2 \log n\Big) \ge \kappa_p.$$

Proof. We approximate μ_u^* by a compactly supported admissible measure θ (Step 1), place one atom at each $1/n$-quantile of θ (Step 2), and show that the resulting single term of Heine's sum loses only $O(n \log n)$ against $n^2 \mathcal{J}_p(\theta)$ (Step 3). The weights cost only $O(n \log n)$ (Step 4).

Fix $\eta > 0$.

Step 1: a compactly supported admissible approximant. By Theorem 1.3(i), μ_u^* has a continuous positive density $\rho^* \le 1$ on $(0,\infty)$. For $U > 0$ let $\mu_U := \mu_u^*|_{(0,U]}$, of mass m_U. By the proof of Theorem 1.3(iii), $|\log|g(u) - g(v)||$ is $\mu_u^* \otimes \mu_u^*$-integrable, so by dominated convergence $\mathcal{J}_p(\mu_U) \to \mathcal{J}_p(\mu_u^*) = \kappa_p$ and $m_U \to 1$ as $U \to \infty$. Let $\theta := (\mu_U)_{1/m_U}$ (notation of Lemma 3.1). Then θ has mass 1, density $\rho^*(um_U) \in [d_U, 1]$ on its support $(0, U']$, where $U' := U/m_U$ and $d_U := \min_{(0,U]} \rho^* \in (0,1]$, and density 0 beyond. So $\theta \in \mathcal{A}$, and by Lemma 3.1

$$\mathcal{J}_p(\theta) = m_U^{-2} \mathcal{J}_p(\mu_U) + p \log m_U \ge \kappa_p - \eta$$

for U large enough, which we fix. Constants C_U and $O_U(\cdot)$ below depend on p and U only (in Step 4 also on c_1 and s).

Step 2: the quantile configuration. Let F be the distribution function of θ; it is continuous and strictly increasing on $[0,U']$. For $1\le i\le n$ let $B_i=[l_i,r_i]:=[F^{-1}(\frac{i-1}{n}),F^{-1}(\frac{i}{n})]$, $q_i:=F^{-1}(\frac{i-1/2}{n})$, $k_i:=\lceil nq_i\rceil$ and $u_i:=k_i/n\in[q_i,q_i+\frac{1}{n})$.

- Since $\theta\le$ Leb and $\theta(B_i)=\frac{1}{n}$: $|B_i|\ge\frac{1}{n}$, so $l_i\ge\frac{i-1}{n}$ and $r_i\ge\frac{i}{n}$; also $r_i-q_i\ge\frac{1}{2n}$ and $q_i-l_i\ge\frac{1}{2n}$. Hence $q_{i+1}-q_i\ge\frac{1}{n}$, so the k_i are distinct, $k_1\ge1$, and $u_j-u_i\ge\frac{1}{n}$ for $i<j$. Moreover, for $i\le n-1$, $u_i<q_i+\frac{1}{n}\le r_i+\frac{1}{2n}\le r_{i+1}$.
- Since the density is $\ge d_U$ on $(0,U']$: $|B_i|\le\frac{1}{d_Un}$, so $r_i\le\frac{i}{d_Un}$, and $u_j\le r_j+\frac{1}{2n}\le\frac{j+1}{d_Un}$.
- $S^*:=\{k_1<\cdots<k_n\}\subset[1,U'n+1]$.

Step 3: energy comparison. Write $F(u,v)=\log|g(u)-g(v)|$ and, for blocks B,B', put $\langle F\rangle_{B\times B'}:=n^2\iint_{B\times B'}F\,\mathrm{d}\theta\,\mathrm{d}\theta$; this is an average, since $\theta(B_i)=\frac{1}{n}$. Then

$$n^2\mathcal{J}_p(\theta)=\sum_{i\ne j}\langle F\rangle_{B_i\times B_j}+\sum_i\langle F\rangle_{B_i^2}.$$

For $i<j$, $u\in B_i$ and $v\in B_j$ we have $u\le v$ and

$$F(u,v)=-p\log u+h(u/v),\qquad h(r):=\log(1-r^p),$$

with h decreasing on $(0,1)$. We show that $F(u_i,u_j)\ge\langle F\rangle_{B_i\times B_j}-\mathrm{err}_{ij}$ with $\sum_{i<j}\mathrm{err}_{ij}=O_U(n\log n)$. We use repeatedly that for $0<r<1$,

$$\min(1,p)(1-r)\le1-r^p\le\max(1,p)(1-r). \tag{5.1}$$

(a) *The term $-p\log u$.* The point value minus the average is $-p\langle\log(u_i/u)\rangle_{B_i}$, and $u_i\le r_{i+1}$, $u\ge l_i$ on B_i. For $i\ge2$, $\log(r_{i+1}/l_i)\le(r_{i+1}-l_i)/l_i\le\frac{2/(d_Un)}{(i-1)/n}$, so the loss is at most $2p/(d_U(i-1))$. For $i=1$, Lemma 5.1 (with $\varphi=\log(1/u)$ and $\lambda=\theta|_{B_1}$, $\delta=1/n$) gives $\langle\log(1/u)\rangle_{B_1}\le n\int_0^{1/n}\log\frac{1}{u}\,\mathrm{d}u=\log n+1$, and $\log u_1\le\log r_2\le\log\frac{2}{d_Un}$; so the loss is at most $p(\log\frac{2}{d_U}+1)$. Summing over $j>i$ and then over i gives $O_U(n\log n)$.

(b) *The term h, for $j\ge i+3$.* On $B_i\times B_j$, $u/v\ge a:=l_i/r_j$, so the average of $h(u/v)$ is at most $h(a)$. At the point, $u_i/u_j\le b:=r_{i+1}/l_j$, and $b<1$ because $l_j=r_{j-1}\ge r_{i+2}>r_{i+1}$; so $h(u_i/u_j)\ge h(b)$. The loss is at most $h(a)-h(b)=\log\frac{1-a^p}{1-b^p}\le\frac{b^p-a^p}{1-b^p}$. By (5.1), $b^p-a^p=b^p(1-(a/b)^p)\le\max(1,p)(1-a/b)$.

If $b\le\frac12$, then $1-b^p\ge1-2^{-p}$, and $1-\frac{a}{b}\le\big(1-\frac{l_i}{r_{i+1}}\big)+\big(1-\frac{l_j}{r_j}\big)\le\frac{2/(d_Un)}{(i+1)/n}+\frac{1/(d_Un)}{j/n}\le\frac{3}{d_U(i+1)}$, so the loss is at most $C_U/(i+1)$.

If $b>\frac12$, then by (5.1), $1-b^p\ge\min(1,p)(1-b)=\min(1,p)\frac{l_j-r_{i+1}}{l_j}\ge\min(1,p)\frac{j-i-2}{nl_j}$, since the $j-i-2$ blocks $B_{i+2},\dots,B_{j-1}$ each have length $\ge\frac1n$. Also $1-\frac{a}{b}=\frac{b-a}{b}\le2(b-a)$ and $b-a\le\frac{r_{i+1}-l_i}{l_j}+\frac{l_i}{r_j}\cdot\frac{r_j-l_j}{l_j}\le\frac{3}{d_Unl_j}$, so the loss is at most $C_U/(j-i-2)$.

Summing, $\sum_{j\ge i+3}\big(C_U/(i+1)+C_U/(j-i-2)\big)=O_U(n\log n)$.

(c) *The term h, for $j\in\{i+1,i+2\}$.* The average of h is ≤0 since $h\le0$. At the point, by (5.1), $1-(u_i/u_j)^p\ge\min(1,p)\frac{u_j-u_i}{u_j}\ge\min(1,p)\frac{1/n}{(j+1)/(d_Un)}=\frac{\min(1,p)d_U}{j+1}$. So the loss is at most $\log(j+1)+C_U$, and there are at most $2n$ such pairs: $O_U(n\log n)$.

(d) *Diagonal blocks.* Since $|g(u)-g(v)| \le g(\min(u,v))$, we have $\langle F\rangle_{B_i^2} \le p\langle \log(1/\min(u,v))\rangle_{B_i^2}$. For $i \ge 2$, $\min(u,v) \ge l_i \ge \frac{i-1}{n}$, giving $\le p\log\frac{n}{i-1}$. For $i=1$ and $n \ge 1/d_U$ we have $B_1 \subset (0,1]$, so $\log\frac{1}{\min(u,v)} \le \log\frac{1}{u} + \log\frac{1}{v}$, and Lemma 5.1 gives $\le 2p(\log n+1)$. In total

$$\sum_i \langle F\rangle_{B_i^2} \le p\big((n-1)\log n - \log(n-1)!\big) + 2p(\log n+1) = O(n).$$

Combining (a)–(d),

$$\begin{aligned}\sum_{i\ne j}\log|g(u_i)-g(u_j)| &= 2\sum_{i<j}F(u_i,u_j)\\ &\ge n^2\mathcal{J}_p(\theta) - \sum_i \langle F\rangle_{B_i^2} - O_U(n\log n)\\ &\ge n^2\mathcal{J}_p(\theta) - O_U(n\log n).\end{aligned}$$

Step 4: conclusion. By Lemma 2.1, $D_n(\sigma_0) \ge w(S^*)V(S^*)$. Here $\log w(S^*) \ge n\log c_1 - |s|\,n\log(U'n+1)$, and by (4.1) and Step 3,

$$\log D_n(\sigma_0) \ge -pn(n-1)\log n + n^2(\kappa_p-\eta) - O_U(n\log n).$$

Let $n\to\infty$ and then $\eta\to 0$. □

6. Proof of Theorem 1.1 and Corollary 1.2

Theorems 4.3, 5.2 and 1.3(iii) together give Theorem 1.1 for $x_k = k^{-p}$. This section adds the constant C and general atoms $f(k)$, by Lemma 2.3, and derives Corollary 1.2 from Lemma 2.4.

Proof of Theorem 1.1. Pure powers. Let $\sigma_0 = \sum_k w_k\delta_{k^{-p}}$ with weights satisfying (1.4). By Theorems 4.3, 5.2 and 1.3(iii) ($E_p = \kappa_p$), $\log D_n(\sigma_0) = -pn^2\log n + \kappa_p n^2 + o(n^2)$. By Lemma 2.2(iii), the measure $\sum_k w_k\delta_{Ck^{-p}}$ has $\log D_n$ larger by $n(n-1)\log C = n^2\log C + O(n)$.

General atoms. Let f satisfy (1.3) and put $h(\xi) := C\xi^{-p}$. Since f' is continuous, $\xi^{p+1}|f'(\xi)|$ is bounded below by a positive constant, and $\xi^{p+1}f'(\xi) \to -pC < 0$, we have $f' < 0$ on $[1,\infty)$; f is bounded because $|f'(\xi)| \le C'\xi^{-p-1}$ is integrable. The ratio $\rho = f'/h' = \xi^{p+1}f'(\xi)/(-pC)$ is bounded above and below by positive constants and tends to 1. All determinants involved are positive by Lemma 2.2(iv), since $w_k > 0$ for all k and f, h are injective. Lemma 2.3 gives $\log D_n(\sigma) = \log D_n(\sum_k w_k\delta_{Ck^{-p}}) + o(n^2)$. □

Proof of Corollary 1.2. By Lemma 2.4, $\log D_n = n\log m_0 + 2\sum_{j<n}(n-j)\log b_j$, so

$$\frac{2}{n^2}\sum_{j=1}^{n-1}(n-j)\log(j^p b_j) = \frac{1}{n^2}\Big(\log D_n - n\log m_0 + 2p\sum_{j=1}^{n-1}(n-j)\log j\Big).$$

By Theorem 1.1 and Lemma 2.5, the bracket equals $-pn^2\log n + (\kappa_p + \log C)n^2 + pn^2\log n - \frac{3p}{2}n^2 + o(n^2) = \log\big(c(p)C\big)n^2 + o(n^2)$. □

Remark 6.1. Theorem 1.1 gives the error $o(n^2)$ only. For pure powers the proof gives no rate, because the rate of $\mathcal{J}_p(\theta)\to\kappa_p$ in Step 1 of Theorem 5.2 and the factor $A^2/(4p\log M)$ in Theorem 4.3 are not quantified.

7. Hankel determinants of zeta values

Corollary 1.4 is Theorem 1.1 applied to one measure, which we identify first. We then compare it with Monien's form for $a = 1$ (Remark 7.1) and, for $a = 2$, with the closed forms of Baricz and Štampach, which give the same constant with the better error $O(n \log n)$ (Remark 7.2).

Proof of Corollary 1.4. Since $2a + b > 1$, every entry $\zeta(a(i + j) + b)$ with $i, j \geq 1$ is a convergent Dirichlet series, and

$$\zeta\big(a(i+j)+b\big) = \sum_{k \geq 1} k^{-(2a+b)} \big(k^{-a}\big)^{(i-1)+(j-1)}.$$

Re-indexing $i, j \mapsto i - 1, j - 1$, the determinant is $D_n(\sigma)$ for $\sigma = \sum_k k^{-(2a+b)} \delta_{k^{-a}}$. Apply Theorem 1.1 with $p = a$, $f(\xi) = \xi^{-a}$ (so $C = 1$ and $\xi^{a+1} f'(\xi) \equiv -a$), and $w_k = k^{-(2a+b)}$, for which $\sum_k w_k < \infty$ and $w_k \geq k^{-s}$ with $s = 2a + b$. The special cases follow from $c(1) = \frac{1}{2}$ and $c(2) = \frac{\pi^2}{16}$. □

For example, $\frac{3a}{2} + \log c(a)$ equals 0.806853, 2.516871, 4.127795, 5.696954 (rounded to six decimals) for $a = 1, 2, 3, 4$ (Table 1).

Remark 7.1 ($a = 1$). Monien's $H_n^{(r)}[\zeta] = \det_{1 \leq i,j \leq n} \zeta(i + j + r)$ is the case $a = 1$, $b = r$ of Corollary 1.4, so $\log H_n^{(r)}[\zeta] = -n^2(\log 2n - \frac{3}{2}) + o(n^2)$ for every real $r > -1$. For $r = 0, 1$ this is Monien's weak asymptotic form [9], read with an $o(n^2)$ error term, as the expansion from Zagier's numerics quoted there suggests; Monien's "$\sim$" by itself asserts only the leading term.

Remark 7.2 ($a = 2$: comparison with closed forms). Baricz and Štampach [1] evaluate Hankel determinants of the Rayleigh functions of order $2m$, which we denote by $R_{2m}(\nu)$ (numbering of the arXiv version of [1]). For $j \in \{0, 1, 2\}$ let $H_n^{(j)}(\nu)$ be the $n \times n$ matrix with (i, i') entry $R_{2(i+i')+2j-2}(\nu)$. By [1, Cor. 3], for $n \in \mathbb{N}$, $\nu \notin -\mathbb{N}$ and $j \in \{0, 1\}$,

$$\det H_n^{(j)}(\nu) = 2^{-2n(n+j)} \prod_{k=1}^{2n+j-1} (\nu + k)^{k-2n-j},$$

and by [1, Rem. 5], $\det H_n^{(2)}(\nu) = 2^{-2n(n+2)}(n+1)(n+\nu+1) \prod_{k=1}^{2n+1} (\nu+k)^{k-2n-2}$. Moreover $R_{2m}(\frac{1}{2}) = \zeta(2m)/\pi^{2m}$ [1, (22)]. Taking $\nu = \frac{1}{2}$ and rescaling rows and columns by powers of π yields closed forms for $\det_{1 \leq i,i' \leq n} \zeta(2(i + i') + b)$ with $b = 2j - 2 \in \{-2, 0, 2\}$. For example, for $b = 0$,

$$\log \det_{1 \leq i,i' \leq n} \zeta\big(2(i+i')\big) = 2n(n+1) \log \frac{\pi}{2} + \sum_{k=1}^{2n} (k - 2n - 1) \log\big(k + \tfrac{1}{2}\big).$$

These three closed forms agree with direct determinants for $1 \leq n \leq 30$, in ball arithmetic, to relative difference at most $4 \cdot 10^{-152}$. The asymptotics of the closed form are elementary: with $m = 2n + 1$, $\sum_{k<m} (m - k) \log(k + \frac{1}{2}) = \frac{m^2}{2} \log m - \frac{3}{4} m^2 + O(m \log m)$ (as in Lemma 2.5), which gives

$$\log \det_{1 \leq i,i' \leq n} \zeta\big(2(i+i')\big) = -2n^2 \log n + \big(3 + \log \tfrac{\pi^2}{16}\big) n^2 + O(n \log n),$$

in agreement with Corollary 1.4 for $a = 2$, $b = 0$, and with a better error term. The closed forms for $j = 0$ and $j = 2$ differ from the case $j = 1$ only by the power of

2 ($2^{-2n(n+j)}$), the power of π (exponent $2\sum_{i\le n}(2i+j-1)$), the range $2n+j-1$ of the product (with $m=2n+j$ in place of $2n+1$), and, for $j=2$, the factor $(n+1)(n+\frac{3}{2})$. These changes alter the right-hand side only by $O(n\log n)$. Hence the same computation gives $-2n^2\log n+(3+\log\frac{\pi^2}{16})n^2+O(n\log n)$ for $b=-2$ and $b=2$ as well. Thus for $a=2$, $b\in\{-2,0,2\}$, Corollary 1.4 (with error $O(n\log n)$) also follows from [1]; the asymptotic statement is not made there.

8. Formal verification

Theorem 1.1, Corollary 1.2 and Corollary 1.4 have been formalised in the Lean 4 proof assistant (version 4.34.1) with the Mathlib library, with the hypotheses as stated in Section 1; in Corollary 1.4, ζ is Mathlib's Riemann zeta function. The formalisation contains no unproved steps and uses only Lean's standard axioms (`propext`, `Quot.sound`, `Classical.choice`).

It also covers the parts of Theorem 1.3 that the proof of Theorem 1.1 uses: admissibility of μ_u^* and its saturation on $(0,u_{\mathrm{sat}}]$ (but not the incomplete-beta formula for ρ^*), the potential of μ^* (part (ii)), the value $\mathcal{J}_p(\mu_u^*)=\kappa_p$, the inequality $-\iint\log|x-y|\,\mathrm{d}\nu\,\mathrm{d}\nu\ge 0$ of Lemma 3.3 (not the Fourier identity and not the equality case), and the bound $\mathcal{J}_p(\mu)\le\kappa_p$ for $\mu\in\mathcal{A}$ with bounded support and integrable energy, which is the case the proof uses. Uniqueness of the maximiser is not formalised.

In a few places the formal proof takes a different route from the one written here; for example, it proves Lemma 3.2 with log-sine integrals instead of Fourier series, and Lemma 3.3 with a Gaussian kernel instead of the Fourier transform. The Lean sources are part of the supplementary material (see the data and code availability statement).

9. Numerical illustrations

The tables in this section illustrate the theorems; they are not used in any proof. Columns 2–4 of Table 1, and the four values of $\frac{3a}{2}+\log c(a)$ quoted in Section 7, are correctly rounded (ball arithmetic); all other entries are floating-point.

Constants and the maximiser. Table 1 lists $c(p)$, u_{sat} and κ_p, together with a Monte Carlo estimate of $\mathcal{L}(\mu^*)$ obtained by sampling μ^* directly (independently of the closed-form value). The density ratio $\mathrm{d}\mu^*/\,\mathrm{d}\lambda_p$ evaluated by quadrature equals 1 for $x\ge 4c$ to four digits (one exception at $p=0.5$, $x=4000c$, where the quadrature returns 0.9862; the closed form of Theorem 1.3(i) gives exactly 1 there) and is strictly below 1 for $x<4c$.

p	$c(p)$	$u_{\mathrm{sat}}=2I(p)/\pi$	κ_p	Monte Carlo $\mathcal{L}(\mu^*)$
0.5	0.408248	0.3750	−0.145880	-0.1442 ± 0.0021
1	0.500000	0.5000	0.806853	0.8054 ± 0.0027
1.5	0.566272	0.5798	1.681320	1.6818 ± 0.0034
2	0.616850	0.6366	2.516871	2.5208 ± 0.0042
3	0.689213	0.7132	4.127795	4.1281 ± 0.0058
4	0.738565	0.7628	5.696954	5.6884 ± 0.0075
6	0.801586	0.8235	8.778837	8.7781 ± 0.0109

Table 1. Constants of Theorem 1.3. Columns 2–4 are correctly rounded; the last column is a floating-point Monte Carlo estimate.

The averaged law. For $x_k = k^{-p}$ and $w_k = k^{-s}$ truncated to $k \le K$, the recurrence coefficients b_j were computed in extended precision, and $S_n := \frac{2}{n^2}\sum_{j<n}(n-j)\log(j^p b_j) - \log c(p)$ is shown in Table 2; by Corollary 1.2, $S_n \to 0$. Doubling K changes the tabulated values by at most $5.4\cdot10^{-4}$ ($p = 1.5$; this is about 20% of the entry at $n = 1600$, which is therefore only indicative), $2.8\cdot10^{-5}$ ($p = 3$), $1.4\cdot10^{-6}$ ($p = 4$) and $7\cdot10^{-8}$ ($p = 6$). For $p = 0.5$ the atoms decay too slowly for the truncation used (K-doubling changes S_n by 10^{-2}), so that case is not illustrated. The values are consistent with an error $O(n\log n)$ in Theorem 1.1 for pure powers (compare Remark 6.1). For the zeta weights of Corollary 1.4 with $(a,b) = (3,1)$ and $(4,1)$ the same statistic is $-0.5544, -0.2181, -0.0750$ and $-0.7452, -0.2922, -0.1004$ at $n = 25, 100, 400$. The identification of $\det\zeta(a(i+j)+b)$ with D_n of $\sum_k k^{-(2a+b)}\delta_{k^{-a}}$ was checked to 12 digits for $(a,b) \in \{(3,1),(4,1),(3,2)\}$ and $n \in \{4,8,12\}$.

p	s	K	$n=25$	50	100	200	400	1600
1.5	1.5	50000	−0.03377	−0.02370	−0.01531	−0.00942	−0.00567	−0.00267
3	3	20000	−0.17266	−0.11244	−0.06962	−0.04161	−0.02423	−0.00782
4	4	6000	−0.27151	−0.17494	−0.10757	−0.06399	−0.03714	–
6	6	2000	−0.46811	−0.29944	−0.18324	−0.10863	–	–

TABLE 2. The averaged statistic S_n (floating-point).

Closed forms for $a = 2$. For the three closed forms of Remark 7.2 ($b = -2, 0, 2$) the quantity $(\log\det + 2n^2\log n)/n^2 - \kappa_2$ equals $-1.33\cdot10^{-4}$, $-3.48\cdot10^{-4}$ and $-5.63\cdot10^{-4}$ at $n = 10^5$, and $(n/\log n)$ times it lies between -4.90 and -1.15 for $n \in \{10^2, 10^3, 10^4, 10^5\}$, in line with the error $O(n\log n)$ proved there.

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Data and code availability. The scripts that produced the tables, their outputs, the ball-arithmetic checks and the Lean sources (theorems `paper1_thm11`, `paper1_cor12` and `paper1_cor14`) are available at https://github.com/speagleg/jacobi-asymptotics (tag `v1.0-paper1` [placeholder: tag to be created at submission]).

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